

Nikolai Khokhlov, PhD

Physicist & R&D Engineer

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Summary

Ph.D. in Physics with over 10 years of experience in academia and government labs, specializing in Experimental Optics, Photonics and Femtomagnetism. Proven track record of translating high-level research challenges into robust automated setups and executing projects within a €150K+ budget. Cross-functional researcher focused on optimizing experimental workflows and delivering high-impactful results, experienced in team-leading and collaboration with international groups.

Optical lab experience: free-space optics | setups ultrafast laser-based setups | optical spectroscopy microscopy and imaging

Software: Python, MATLAB, LabView, OOMMF, Boris Spintronics, C#

Professional competencies: Project management & workflow optimization | Team leadership & mentorship | Grant writing & budgeting | Design of educational courses & training

Professional Experience

Postdoc | Radboud University

Nijmegen, Netherlands, 2022 – current

- Lead end-to-end experimental projects in ultrafast laser-matter interactions and femtomagnetism, obtaining results, that earned recognition through 5 high-impact publications (Nature Physics, Phys. Rev. Letters, etc), >10 conference talks and 2 PhD defenses.
- Designed and assembled optical dual-pump-probe setup with combination of sub-picosecond and sub-micron resolutions at cryogenic temperatures.
- Spearheaded the migration of experimental setups automation from LabVIEW to Python, accelerating software development and data analysis pipelines up to 75% for a team of Python-ready students.
- Direct daily workflows and technical mentoring for a team of PhD candidates, optimizing laboratory resource allocation to ensure on-schedule project delivery.

Senior Research Fellow | Ioffe Institute

St. Petersburg, Russia, 2017 – 2022

- Managed spin waves research line, successfully securing and allocating a €150K+ budget for infrastructure development and operations.
- Planned and supervised the construction of two experimental lab rooms for femtosecond optical spectroscopy and microscopy, managing full lifecycle equipment procurement and vendor vetting.
- Engineered custom automation software for ultrafast optical spectroscopy systems, increasing data acquisition and processing efficiency up to 70%.
- Supervised and mentored a team of 2 PhD and 5 MSc students, guiding them through successful on-schedule project delivery.

Research Fellow | Russian Quantum Center

Moscow, Russia, 2013 – 2017

- Execution of experimental projects in magneto-photonics and plasmonics, delivering high-impact publications (Optics Letters, Optics Express, etc).
- Engineered instrumentation interfaces and control architectures using Matlab and LabView for

- complex optical measurements.
- Planned, vendors vetted and supervised the construction of lab room for magneto-optical spectroscopy.

Additional Professional Experience (part-time)

Invited Associate Professor | ITMO University St. Petersburg, Russia, 2018-2022

- Engineered and delivered specialized technical training curricula in Micromagnetics and Magnetophotonics for MSc students.

Video Engineer | Innovative Medical Technologies Ltd Moscow, Russia, 2016 – 2020

- Maintained network infrastructure during medical broadcasting conferences, ensuring 99.9% system uptime during live surgery.
- Designed and deployed robust hardware-software network infrastructure for low-latency video streaming and multi-room synchronization across surgical environments.

Additional Info

Languages: English, Russian (Native), Dutch (A1)

Education:

Ph.D. in Physics

Moscow State University, Russia, 2011 – 2015

M.Sc. Equivalent (Specialist)

Moscow State University, Russia, 2005 – 2011

Hobbies: Fencing (épée), hiking/backpacking (guide license), roller skating (coach license), skiing, climbing, painting